

Progress Report

- Increment 2 -

Group 16

1) Team Members

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2) Project Title and Description

The Market Signal & Research Platform (MSRP) is a full-stack financial analytics web application designed to help users analyze stock market data and evaluate trading strategies. The system allows users to retrieve historical stock data, compute technical indicators such as Simple Moving Average (SMA) and Relative Strength Index (RSI), and run long-only backtests on rule-based strategies.

3) Accomplishments and overall project status during this increment

During this development increment, the team achieved the following:

- Implemented core analytical features, including technical indicators and backtesting functionality
- Developed a working user interface for interacting with stock data and strategies
- Added validation guards and resolved authentication mismatches
- Introduced unit testing and smoke testing to improve system reliability
- Expanded functionality with symbol search and additional financial metrics

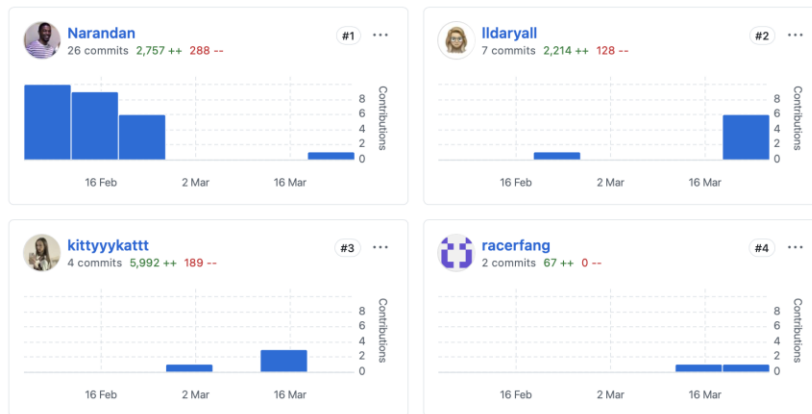
At this stage, MSRP is a functional prototype capable of performing end-to-end analysis workflows.

4) Challenges, changes in the plan and scope of the project and things that went wrong during this increment

The primary challenge during this increment was team coordination. Due to conflicting schedules, it was difficult to consistently organize meetings and maintain communication.

5) Team Member Contribution for this increment

- a) the **progress report** – **Katya Serechenko**
- b) the **requirements and design document** – **Logan Bramblett**
- c) the **implementation and testing document** – **Narandan Miller**
- d) the **source code**
 - **Katya Serechenko**: Fixed issues in Github issue tracker. Overall: built and integrated the front-end with the backend. Added unit and smoke tests as well as fixed minor issues.
 - **Narandan Miller**: Populated issue tracker.
 - **Dascia Pylpenko**: Added the back tester, metrics and strategies. Fixed the auth mismatch. Added validation guard. Add symbol search, Bollinger Bands, EMA, and Sharpe ratio
 - **Logan Bramblett**: Identified need for main API integration file
- e) the **video** – **Dascia Pylpenko**



6) Plans for the next increment

We plan to hold a meeting in a few days to discuss the next sprint iterations. Overall, we do need some polishing on the UI side. We potentially also want to add ML/AI integration, as one of our members – Dascia Pylpenko is certified and wants to put her skills to use. Additionally, some of us would like to deploy this project via vercel/supabase in order to make this a long-lasting project.

7) Stakeholder Communication

Dear Stakeholders,

We are pleased to share an update on the Market Signal & Research Platform (MSRP).

At this stage, we have successfully delivered a functional integration between the user interface and backend systems. This enables users to interact with the platform to retrieve historical market data, apply key technical indicators, and execute rule-based back testing strategies.

In addition to feature development, we have focused on improving system reliability. This includes implementing validation safeguards, resolving integration inconsistencies, and introducing unit and smoke testing to ensure stability as the platform evolves.

Looking ahead, our priorities for the next phase include refining the user interface to enhance usability and visual clarity, as well as preparing the platform for deployment to a cloud-based environment. This will allow for broader accessibility and long-term scalability.

We are also exploring the integration of machine learning capabilities to provide more advanced insights and predictive analytics. This initiative is still in the planning stage, and we look forward to discussing potential directions and use cases with you to ensure alignment.

Thank you for your continued support and engagement.

Best regards,
The MSRP Team

8) Link to video

https://youtu.be/0UbBxS7_lxI